Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

JAVA ASSIGNMENT

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DATE OF SUBMISSION:28/03/24

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Experiment No. - 7 Java

Aim:- To implement constructors and constructor overloading.

Description :-

(i) Type of constructors:-

- Default constructor : A constructor that has no parameters is known as default constructor. It is invisible. If a constructor is written with no argument, the constructor is written with no argument, the compiler doesnt create default constructor is written, it is taken out. It is being overloaded and called a parameterized constructor, which can't change default.

- Parameterized Constructor :- A constructor having parameters is called parameterized constructor. It is used to initialize field of the class with our own value.

- Copy constructor : It is passed with another object which copies the data available from the passed object to the newly created object.

(ii) Properties of a Constructors :-

- It has no return type. If a returned type is declared, JVM considers it a normal method.

- Access Specifier can be private, default, protected, public

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- A constructor can invoke another constructor of the same class. This can be done by using 'this()' statement.
- A constructor has the same name as the class.
- A constructor overloading is supported in Java, i.e. a class can have multiple constructors with different parameters.

(iii) Uses of a constructor

- Object Initialization :- Constructors initialize the state of an object. They set initial value for instance variables.
- Creation of Objects : Constructors are called to create instances of a class using the 'new' keyword.
- Constructor can be overloaded to provide multiple ways to initialize objects, allowing flexibility in object creation.
- Inheritance : Constructors facilitate inheritance by initializing the state of both the subclass and superclass. Keyword 'super()' can be used.

(iv) Constructor overloading :—

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- Constructor overloading in Java is a form of static polymorphism, also known as compile-time polymorphism.
- It allows a class to have multiple constructors with different parameter lists.
- The choice of which constructor to call is determined at compile-time based on the arguments provided during object creation.

Conclusion:-

Constructors and constructor overloading can be useful while writing a code in Java.

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1. To implement Inheritance:-

Q. WAP to demonstrate the role of Constructors in Inheritance in the following class diagram.

Code:-

```
Class A
{
    A ()
    {
        Super ();  /* this Super() method will class Parent
class Object which is default parent class of user defined
class */
        System.out.println ("In the Constructor of Class A");
    }
}
Class B extends A
{
    B()
    {
        Super ();
        System.out.println ("In the Constructor of Class A");
    }
    B (int a, int b)
    {
        this ();
        System.out.println ("Addition of " + a + " + b + " =
```

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```
    " +(a+b) );
    }
}
class C extends B
{
    C()
    {
        super(10,20);
        // super() method is to call super class default or
        parameterized constructor.
        System.out.println("In the Constructor of Class C");
    }
    C(int a)
    {
        this();//this() method is to call overloaded constructor
        of same class
        System.out.println("Value of a= " +a);
    }
}

class ConstructorInInheritance
{
    public static void main (String args[])
    {
        C obj = new C(20);
    }
}
```

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Experiment No: 7

Q1] WAP to count the no. of objects created of a class using constructors.

Code:

```
public class Constructor {
static int count=0;
public Constructor(){
count++;
}
static void display(){
System.out.println("The number of objects are: "+count);
}
public static void main(String[] args) {
Constructor obj1=new Constructor();
Constructor obj2=new Constructor();
Constructor obj3=new Constructor();
display();
}
}
```


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Output :

```
Microsoft Windows [Version 10.0.19045.4170]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lenovo>cd C:\java program

C:\java program>javac Constructor.java

C:\java program>java Constructor
The number of objects are: 3

C:\java program>_
```


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Q2] WAP to display area of square and rectangle using the concept of overloaded constructor (use parameterized, non-parameterized and copy constructor).

Code:

```
import java.util.*;
public class Shape {
int s;
float l,b;
public static void main(String[] args) {
Shape s=new Shape();
Shape s1=new Shape(6);
Shape s2=new Shape(2.1f,5.1f);
Shape s3=new Shape(s2);
}
Shape(){
Scanner sc=new Scanner(System.in);
System.out.println("Enter the side of the square: ");
int s=sc.nextInt();
System.out.println("The area of square: "+(s*s));
}
Shape(int a){
s=a;
System.out.println("The area of square= "+(s*s));
}
```


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```
Shape(float x,float y){
l=x;
b=y;
System.out.println("Area of the recttangle= "+(l*b));
}
Shape(Shape s2){
l=s2.l;
b=s2.b;
System.out.println("Area of rectangle= "+(l*b));
}
}
```

Output:

```
Microsoft Windows [Version 10.0.19045.4170]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lenovo>cd C:\java program

C:\java program>javac Shape.java

C:\java program>java Shape
Enter the side of the square:
5
The area of square: 25
The area of square= 36
Area of the recttangle= 10.709999
Area of rectangle= 10.709999

C:\java program>_
```


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CONCLUSION:

• We learnt the implementation of an array of objects and the use to pass by value
and pass by
reference method
• We also understood how to create objects that act like pointers in Java
• We also saw the use of 'this' reference

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